

COSC 594/650

Dual Method and Convergence

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Slater's condition and sensitivity

$$\begin{array}{ll}\underset{x}{\text{minimize}} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & x \in X\end{array}$$

- GEOMETRIC INTERPRETATION OF DUALITY
- SLATER'S CONDITION AND STRONG DUALITY
- MULTIPLIERS ARE SHADOW PRICES

Problem	Strong duality	KKT conditions	Sensitivity
General	may fail	necessary if $d^* = p^*$	—
Convex + Slater	$d^* = p^*$	necessary & sufficient	$\lambda_i^* = -\partial p^* / \partial u_i$

QUESTION: How fast does the dual method converge?

From the dual problem to an algorithm

Primal problem: $\underset{x}{\text{minimize}} \quad f_0(x) \quad \text{subject to} \quad Ax \leq b$

Lagrangian: $L(x, \lambda) = f_0(x) + \lambda^\top (Ax - b)$

■ TWO STANDING ASSUMPTIONS

- ★ f_0 differentiable and σ -strongly convex – $L(\cdot, \lambda)$ has a unique minimizer
- ★ $A \neq 0$ and $A\bar{x} < b$ for some $\bar{x}$ – Slater point

■ DUAL PROBLEM

$$g(\lambda) = \inf_x L(x, \lambda), \quad d^* = \underset{\lambda \geq 0}{\text{maximize}} \quad g(\lambda)$$

Strong duality: $d^* = p^*$, and both x^* and λ^* are attained

The inner problem and the dual algorithm

Dual function: $g(\lambda) = \inf_x L(x, \lambda), \quad \lambda \geq 0$

■ INNER PROBLEM

$$x(\lambda) = \operatorname{argmin}_x \left(f_0(x) + \lambda^\top (Ax - b) \right)$$

- ★ strong convexity – $x(\lambda)$ is unique
- ★ solved exactly at every iteration – it must be affordable

■ DUAL ALGORITHM

$$\lambda_{k+1} = [\lambda_k + \alpha \nabla g(\lambda_k)]_+, \quad x_k = x(\lambda_k)$$

Dual method: One unconstrained solve per iteration, then one dual update

The dual gradient is the constraint residual

Two multipliers: $\lambda \geq 0$ and $\eta \geq 0$

$$\begin{aligned} g(\eta) &= \inf_x L(x, \eta) \\ &\leq L(x(\lambda), \eta) \\ &= L(x(\lambda), \lambda) + (\eta - \lambda)^\top (Ax(\lambda) - b) \\ &= g(\lambda) + (\eta - \lambda)^\top (Ax(\lambda) - b) \\ &\implies \nabla g(\lambda) = Ax(\lambda) - b \end{aligned}$$

- ★ $Ax(\lambda) - b$ – supergradient of the concave function g
- ★ $[Ax(\lambda) - b]_i > 0$ – violated constraint, and raise the price λ_i

Dual gradient ascent

Initialization: $\lambda_0 \geq 0$, and stepsize $\alpha > 0$

$$\begin{aligned}x_k &= \operatorname{argmin}_x \left(f_0(x) + \lambda_k^\top (Ax - b) \right) \\ \lambda_{k+1} &= [\lambda_k + \alpha (Ax_k - b)]_+\end{aligned}$$

■ PRICE UPDATE

- ★ violation raises λ_i and slack lowers it – floor at zero
- ★ $[\lambda]_+ = \max\{\lambda, 0\}$ – projection onto the orthant, componentwise

■ WHEN IS THIS USEFUL?

- ★ $f_0(x) = \sum_j f_{0j}(x_j)$ – the inner problem splits into n scalar problems

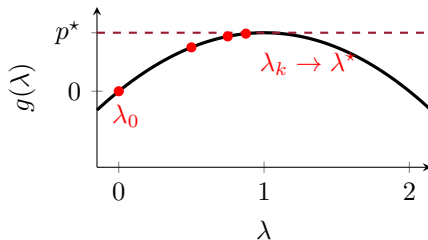
Gradient ascent on g , and projected onto $\lambda \geq 0$

Example: one inequality constraint

$$\begin{array}{ll} \underset{x}{\text{minimize}} & \frac{1}{2} x^2 \\ \text{subject to} & 1 - x \leq 0 \end{array} \quad \begin{array}{l} g(\lambda) = \lambda - \frac{1}{2} \lambda^2 \\ x(\lambda) = \lambda \end{array}$$

$$\alpha = \frac{1}{2} : \quad \lambda_{k+1} = \left[\lambda_k + \frac{1}{2}(1 - \lambda_k) \right]_+ = 1 - 2^{-(k+1)}$$

k	$\lambda_k = x_k$	$[1 - x_k]_+$	$p^* - g(\lambda_k)$
0	0	1	1/2
1	1/2	1/2	1/8
2	3/4	1/4	1/32
3	7/8	1/8	1/128



The dual value improves while every x_k stays infeasible

Smoothness of the dual function

Inner optimality: $\nabla f_0(x(\lambda)) = -A^\top \lambda$, $\nabla f_0(x(\eta)) = -A^\top \eta$

■ STRONG CONVEXITY OF f_0

$$\begin{aligned}\sigma \|x(\lambda) - x(\eta)\|^2 &\leq \langle \nabla f_0(x(\lambda)) - \nabla f_0(x(\eta)), x(\lambda) - x(\eta) \rangle \\ &= -\langle A^\top (\lambda - \eta), x(\lambda) - x(\eta) \rangle \\ &\leq \|A\|_2 \|\lambda - \eta\| \|x(\lambda) - x(\eta)\|\end{aligned}$$

$$\implies \|x(\lambda) - x(\eta)\| \leq \frac{\|A\|_2}{\sigma} \|\lambda - \eta\| \quad - \text{ the primal move is limited}$$

■ LIPSCHITZ DUAL GRADIENT

$$\|\nabla g(\lambda) - \nabla g(\eta)\| = \|A(x(\lambda) - x(\eta))\| \leq \underbrace{\frac{\|A\|_2^2}{\sigma}}_{L_d} \|\lambda - \eta\|$$

One-step inequality for dual gradient ascent

Dual gradient ascent: $\lambda_{k+1} = [\lambda_k + \alpha \nabla g(\lambda_k)]_+, \quad 0 < \alpha \leq 1/L_d$

■ CONCAVITY AND SMOOTHNESS

$$g(\mu) - g(\lambda_{k+1}) \leq \nabla g(\lambda_k)^T (\mu - \lambda_{k+1}) + \frac{L_d}{2} \|\lambda_k - \lambda_{k+1}\|^2$$

■ PROJECTION OPTIMALITY AND SQUARED DISTANCES

$$\begin{aligned} \nabla g(\lambda_k)^T (\mu - \lambda_{k+1}) &\leq \frac{1}{\alpha} (\lambda_{k+1} - \lambda_k)^T (\mu - \lambda_{k+1}) \\ &= \frac{\|\lambda_k - \mu\|^2 - \|\lambda_{k+1} - \mu\|^2 - \|\lambda_{k+1} - \lambda_k\|^2}{2\alpha} \end{aligned}$$

$$g(\mu) - g(\lambda_{k+1}) \leq \frac{\|\lambda_k - \mu\|^2 - \|\lambda_{k+1} - \mu\|^2}{2\alpha}$$

Sufficient ascent of the dual function

One-step inequality: $g(\mu) - g(\lambda_{k+1}) \leq \frac{\|\lambda_k - \mu\|^2 - \|\lambda_{k+1} - \mu\|^2}{2\alpha}$

■ CHOOSE THE CURRENT ITERATE $\mu = \lambda_k$:

$$g(\lambda_k) - g(\lambda_{k+1}) \leq -\frac{\|\lambda_{k+1} - \lambda_k\|^2}{2\alpha}$$

$$g(\lambda_{k+1}) - g(\lambda_k) \geq \frac{\|\lambda_{k+1} - \lambda_k\|^2}{2\alpha} \geq 0$$

■ THE DUAL OPTIMALITY GAP CANNOT INCREASE

$$g(\lambda^*) - g(\lambda_{k+1}) \leq g(\lambda^*) - g(\lambda_k)$$

Convergence analysis for dual gradient ascent

Sufficient ascent: $g(\lambda^*) - g(\lambda_{k+1}) \leq g(\lambda^*) - g(\lambda_k)$

■ COST-TO-GO ESTIMATE

$$\|\lambda_{k+1} - \lambda^*\|^2 \leq \|\lambda_k - \lambda^*\|^2 - 2\alpha [g(\lambda^*) - g(\lambda_{k+1})]$$

$$\Rightarrow \sum_{k=0}^{K-1} [g(\lambda^*) - g(\lambda_{k+1})] \leq \frac{\|\lambda_0 - \lambda^*\|^2 - \|\lambda_K - \lambda^*\|^2}{2\alpha} \leq \frac{\|\lambda_0 - \lambda^*\|^2}{2\alpha}$$

■ BOUND EVERY TERM BY THE LAST ONE

$$K [g(\lambda^*) - g(\lambda_K)] \leq \sum_{k=0}^{K-1} [g(\lambda^*) - g(\lambda_{k+1})]$$

$$\Rightarrow 0 \leq p^* - g(\lambda_K) \leq \frac{\|\lambda_0 - \lambda^*\|^2}{2\alpha K}$$

Primal recovery and a stopping test

Strong convexity of $L(\cdot, \lambda_k)$: x_k minimizes $L(\cdot, \lambda_k)$

$$g(\lambda_k) + \frac{\sigma}{2} \|x_k - x^*\|^2 \leq L(x^*, \lambda_k) \leq f_0(x^*) = p^*$$

$$\implies \|x_k - x^*\| \leq \frac{\|\lambda_0 - \lambda^*\|}{\sqrt{\sigma \alpha k}}$$

■ STOPPING TEST

- ★ $\|[Ax_k - b]_+\| \leq \|A\|_2 \|x_k - x^*\|$ – x_k is infeasible in general
- ★ $\hat{x}_k$ feasible – weak duality certifies accuracy

$$0 \leq f_0(\hat{x}_k) - p^* \leq f_0(\hat{x}_k) - g(\lambda_k) \leq \epsilon$$

Equality constraints: linear convergence

$$\begin{array}{ll} \underset{x}{\text{minimize}} & \frac{1}{2} \|x\|^2 \\ \text{subject to} & Ax = b \end{array} \qquad \begin{array}{l} g(\mu) = -\frac{1}{2} \mu^\top AA^\top \mu - b^\top \mu \\ x(\mu) = -A^\top \mu \end{array}$$

■ NO PROJECTION: μ IS FREE

$$\mu_{k+1} = \mu_k - \alpha (AA^\top \mu_k + b) \quad \implies \quad \mu_{k+1} - \mu^\star = (I - \alpha AA^\top) (\mu_k - \mu^\star)$$

$$\star \alpha = 1/\lambda_{\max}(AA^\top)$$

■ LINEAR RATE

$$\|\mu_k - \mu^\star\| \leq \left(1 - \frac{1}{\kappa}\right)^k \|\mu_0 - \mu^\star\|$$

$$\star \kappa = \lambda_{\max}(AA^\top)/\lambda_{\min}(AA^\top) - \text{dual condition number}$$

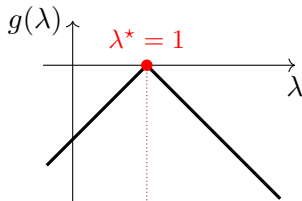
A nonsmooth dual function

Convex, not strongly convex: $x_k \in \operatorname{argmin}_{x \in X} L(x, \lambda_k)$ – may be a set

■ **EXAMPLE** $f_0(x) = -x, \quad x \leq 0, \quad X = [-1, 1]$

$$g(\lambda) = \inf_{|x| \leq 1} (\lambda - 1)x = -|\lambda - 1|$$

- ★ a kink at λ^* – use a supergradient
- ★ $\|Ax_k - b\| \leq G$, and $R = \|\lambda_0 - \lambda^*\|$



■ SUPERGRADIENT STEP

$$\|\lambda_{k+1} - \lambda^*\|^2 \leq \|\lambda_k - \lambda^*\|^2 - 2\alpha [g(\lambda^*) - g(\lambda_k)] + \alpha^2 G^2$$

$$g(\lambda^*) - \max_{0 \leq k < K} g(\lambda_k) \leq \frac{R^2}{2\alpha K} + \frac{\alpha G^2}{2} \quad \alpha = \underline{R/(G\sqrt{K})} \quad \frac{RG}{\sqrt{K}}$$

Dual method and convergence

$$x_k = \operatorname{argmin}_{x \in X} L(x, \lambda_k), \quad \lambda_{k+1} = [\lambda_k + \alpha (Ax_k - b)]_+$$

- THE DUAL GRADIENT IS THE CONSTRAINT RESIDUAL
- STRONG CONVEXITY MAKES THE DUAL SMOOTH
- PRIMAL ITERATES ARE INFEASIBLE ALONG THE WAY

Assumption	Step size	Dual gap	Primal error
σ -strongly convex f_0	$\alpha \leq \sigma / \ A\ _2^2$	$O(1/K)$	$O(1/\sqrt{K})$
$Ax = b, \quad AA^\top \succ 0$	$\alpha = 1/\lambda_{\max}$	$(1 - 1/\kappa)^{2k}$	$(1 - 1/\kappa)^k$
convex only	$\alpha = R/(G\sqrt{K})$	$O(K^{-1/2})$	no general bound

Questions?

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